

EDUVISION DV

Project Report

Higher-Education Analytics and Tableau Dashboard



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Executive Summary

EduVision DV is a completed higher-education analytics project that transforms university ranking and institutional performance data into an integrated Tableau decision-support environment. The completed 2026 merged dataset contains 1,120 universities across 93 countries and 50 fields. It integrates a QS World University Rankings 2026 source dataset containing 1,501 universities and 30 columns with a Times Higher Education source containing 16,713 records covering 2016–2026. The analytical design goes beyond a single ranking position by combining global ranking, research, student, internationalisation, reputation and country-comparison perspectives.

At-a-Glance Project Facts

Area	Completed project fact
Final analytical universe	1,120 universities; 93 countries; 50 fields
QS source	1,501 universities; 30 columns
THE source	16,713 records; 2016–2026
Completed dashboards	University Overview; Research Analytics; Student Analytics; Country Comparison
Analytical worksheets	52 worksheets
Major interaction layer	Year, Region, Country, Focus, Research, Size and Status filters; parameters; actions; reset controls
Primary technology	Python, Pandas, NumPy and Tableau Desktop/Tableau Public workflow as specified in the project materials

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1. Project Background and Business Context

Higher-education decision making is often fragmented across ranking positions, research metrics, student composition and institutional characteristics. EduVision_DV addresses this fragmentation by creating a unified analytical layer that combines ranking data with multidimensional institutional indicators and exposes them through interactive Tableau dashboards.

The project was designed around the practical need to move beyond “who is ranked highest?” toward “why does an institution or country perform the way it does, and how can different stakeholders compare that performance?” This framing is consistent with the project specification, which positions the dashboard suite as a tool for institutional comparison, research analysis, student diversity analysis and country benchmarking.

Problem Statement

- Raw ranking datasets originate from different providers, structures and scoring conventions.
- A single rank does not adequately explain research impact, student environment, internationalisation or country-level performance.
- Users need filterable and comparable views rather than isolated static ranking tables.
- Decision support requires derived KPIs that combine related variables into interpretable measures.

Project Response

- Integrate QS World University Rankings 2026 data with Times Higher Education ranking data.
- Clean and standardize university and country data and normalize ranking metrics.
- Engineer six analytical KPIs.
- Build four interconnected Tableau dashboards supported by 52 analytical worksheets.
- Provide filters, parameters, dashboard actions, ranking-movement calculations, Top-10 logic and reset-filter controls.

2. Objectives, Scope and Intended Users

Objectives

1. Create a comprehensive higher-education analytics environment from multiple global ranking sources.
2. Produce consistent metrics suitable for cross-university and cross-country comparison.
3. Provide research-performance intelligence beyond overall ranking positions.
4. Provide student-scale, student-diversity and internationalisation intelligence.
5. Enable country-level benchmarking and ranking-movement analysis.
6. Package the analysis as a coherent, interactive Tableau workbook for practical decision support.

Intended Users

- Students and prospective students
- Academic researchers
- University administrators
- Policymakers
- Education consultants

Scope Boundary

The completed project is an analytics and visualization solution. It does not claim to replace the source ranking organizations, create an official ranking methodology, or establish causal explanations for why an institution changes rank. The dashboards are designed to support comparative analysis and decision making using the integrated dataset.

3. Data Sources and Final Dataset

Source Dataset Profile

Source	Records / Universities	Fields / Columns	Coverage
QS World University Rankings 2026	1,501 universities	30 columns	2026 ranking dataset
Times Higher Education World University Rankings	16,713 records	Not specified in supplied project summary	2016–2026
Final EduVision_DV merged dataset	1,120 universities	50 fields	2026 final analytical dataset

Interpretation of counts: The 1,501 QS figure, 16,713 THE record figure and 1,120-university final figure are the project facts supplied for this report. The reduction from source records to the final integrated universe is the result of the project's collection, cleaning and integration workflow; the exact row-matching/reconciliation algorithm is not further specified in the supplied summary.

Final Dataset Field Groups

- Identity and segmentation: Rank_QS, Previous_Rank, Name, Region, Size, Focus, Research, Status, Year, Country.
- QS metrics: Academic Reputation Score/Rank, Employer Reputation Score/Rank, Faculty-Student Ratio Score/Rank, Citations per Faculty Score/Rank, International Faculty Score/Rank, International Student Score/Rank, International Students Diversity Score/Rank, International Research Network Score/Rank, Employment Outcomes Score/Rank, Sustainability Score/Rank, QS Overall Score.
- THE metrics: Rank_THE, Student Population, Students to Staff Ratio, International Students, Female to Male Ratio, THE Overall Score, Teaching, Research Environment, Research Quality, Industry Impact, International Outlook.
- Derived metrics: QS_Rank_Normalized, THE_Rank_Normalized, KPI_Global_Ranking_Score, KPI_Research_Impact_Score, KPI_Faculty_to_Student_Ratio, KPI_International_Student_Percentage, KPI_Academic_Reputation_Score, KPI_Research_Productivity_Index.

4. Data Preparation and Integration Methodology

Completed Workflow

1. Data collection
2. Cleaning and transformation
3. University-data integration
4. Ranking normalization
5. KPI engineering
6. Tableau worksheet construction
7. Dashboard development
8. Dashboard integration
9. Testing and validation
10. Documentation and delivery

Data Cleaning and Transformation Activities

- Remove duplicates.
- Standardize university names.
- Standardize country names.
- Normalize ranking metrics.
- Create Tableau-ready datasets.

Integration Logic

The project combines the QS and THE sources into a common analytical structure so that institutional performance can be assessed using both ranking ecosystems and additional indicators. The supplied methodology explicitly calls for

ranking normalization before KPI engineering. The project summary defines the final Global Ranking Score as an average of normalized QS and THE rank scores; the precise mathematical normalization transform is not stated in the supplied materials and is therefore not assumed here.

KPI Readiness

Once the cleaned integrated dataset was prepared, the project derived six KPIs that provide a more decision-oriented view of university performance. These metrics are then exposed in the Tableau workbook for aggregation, comparison, filtering and visualization.

5. KPI Engineering

#	KPI	Formula / Definition	Analytical role
1	Global Ranking Score	Average of normalized QS and THE rank scores	Produces a combined ranking-oriented score across the two ranking sources.
2	Research Impact Score	35% Citations per Faculty + 45% THE Research Quality + 20% Industry Impact	Combines citation intensity, research quality and industry relevance.
3	Faculty-to-Student Ratio	1 / Students-to-Staff Ratio	Transforms the students-to-staff measure into its reciprocal form so that the derived metric follows the naming convention used by EduVision_DV.
4	International Student Percentage	International Students $\times$ 100	Converts the International Students field into a percentage representation as defined in the project.
5	Academic Reputation Score	QS Academic Reputation Score	Carries the QS academic reputation indicator into the EduVision KPI layer.
6	Research Productivity Index	30% Research Quality + 25% Research Environment + 20% Citations per Faculty + 15% International Research Network + 10% Industry Impact	Creates a composite research-productivity measure using five research-related dimensions.

Formula handling: The formulas above are reproduced from the project information supplied for EduVision_DV. This report does not silently add normalization, scaling, capping or missing-value rules that are not stated in the source materials.

Why KPI Engineering Matters

- It converts disparate source indicators into consistent analytical concepts that can be surfaced in Tableau.
- It makes research, student and internationalisation questions easier to answer without repeatedly reconstructing formulas.
- It supports comparison at both institution and country levels through Tableau aggregation.

6. Tableau Architecture and Interaction Design

Workbook Structure

Layer	Project implementation	Purpose
Data layer	Embedded Tableau Hyper extract (as observed in the packaged workbook); integrated 2026 analytical dataset	Supplies the dashboard data.
Analytical layer	52 worksheets; calculated fields; KPI measures; ranking movement; Top-10 logic	Transforms the data into reusable analytical views.
Interaction layer	Filters, parameters, dashboard actions, navigation and reset controls	Allows users to explore, compare and return to a clean state.
Presentation layer	4 integrated dashboards	Communicates findings by decision domain.

Common Filters

- Year
- Region
- Country
- Focus
- Research
- Size
- Status

Parameter and Action Design

- Metric Selector parameter supporting Global Score, Research Impact and Research Productivity is part of the workbook interaction layer.
- Dashboard navigation connects the four main analytical views.
- Calculated fields support ranking movement, Top-10 views and KPI aggregation.
- Reset-filter controls provide a clean interaction state without manually clearing every filter.

Architecture boundary: The report describes the interaction patterns actually identified in the supplied workbook/screenshots. It does not infer additional actions or filters beyond those visible or stated in the project information.

7. Dashboard 7: University Overview

The University Overview dashboard is the portfolio-level entry point. It summarizes the size of the analytical universe, gives a regional benchmark, surfaces leading institutions and provides a high-level view of ranking movement and institutional scale.

Displayed KPI cards

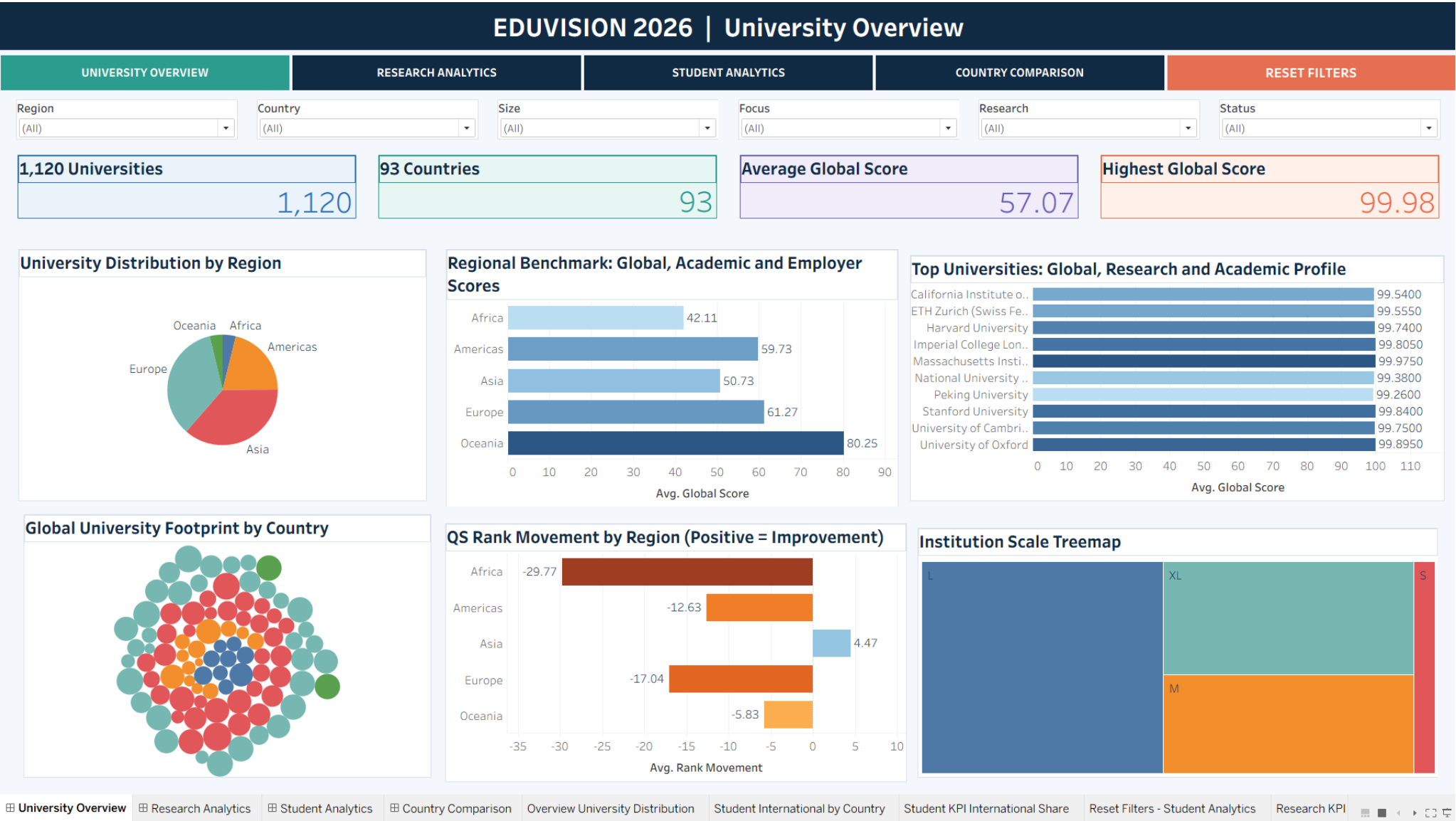
KPI	Displayed value
Universities	1,120
Countries	93
Average Global Score	57.07
Highest Global Score	99.98

Visual components

- University Distribution by Region (pie chart).
- Regional Benchmark: Global, Academic and Employer Scores (regional horizontal bars using the displayed Avg. Global Score measure).
- Top Universities: Global, Research and Academic Profile (Top-10 horizontal bar chart).
- Global University Footprint by Country (packed-bubble style visualization).
- QS Rank Movement by Region (Positive = Improvement).
- Institution Scale Treemap.

Interpretation from the supplied dashboard view

- Oceania has the highest displayed regional average Global Score at 80.25; Africa is lowest at 42.11.
- The displayed regional averages are: Americas 59.73, Asia 50.73 and Europe 61.27.
- The Top-10 global-score view is tightly clustered close to 100, with displayed values ranging from 99.2600 to 99.9750.
- Among the five regional rank-movement values, Asia is the only positive value at +4.47; Africa shows the largest negative movement at -29.77.
- The dashboard uses all-common filters at the top, allowing region/country/institution segmentation before interpreting the charts.



8. Dashboard 8: Research Analytics

The Research Analytics dashboard focuses on the research-production and research-impact side of higher education performance. It combines a composite KPI layer with institution-level and region-level views.

Displayed KPI cards

KPI	Displayed value
Average Research Impact	50.22
Average Research Productivity	46.41
Average Research Quality	59.90
Average Citations per Faculty	33.03

Visual components

- Research Quality vs Citations per Faculty (scatter plot).
- Research Performance by Region (regional Avg. Research Impact).
- Research Activity Footprint by Region (institutional footprint by category).
- Global Research Impact Map (packed-bubble style visualization).
- Top Research Institutions (Top-10 by displayed Avg. Research Impact).
- Research Quality and Network Strength (regional line chart using displayed Avg. Research Quality).

Interpretation from the supplied dashboard view

- Oceania leads displayed regional Research Impact at 75.68, followed by the Americas at 52.32, Europe at 51.54, Asia at 46.46 and Africa at 37.48.
- The regional Research Quality values displayed in the line chart are Africa 56.91, Americas 62.03, Asia 54.93, Europe 61.75 and Oceania 82.20.
- The scatter plot shows a generally upward visual relationship between citations per faculty and research quality, while retaining substantial dispersion, especially toward the lower citation range.
- The Top Research Institutions view contains ten displayed institutions with scores from 96.035 to 99.820 on the shown research-impact measure.
- The research dashboard therefore supports both institutional identification (Top-10) and structural interpretation (regional and citation-quality patterns).

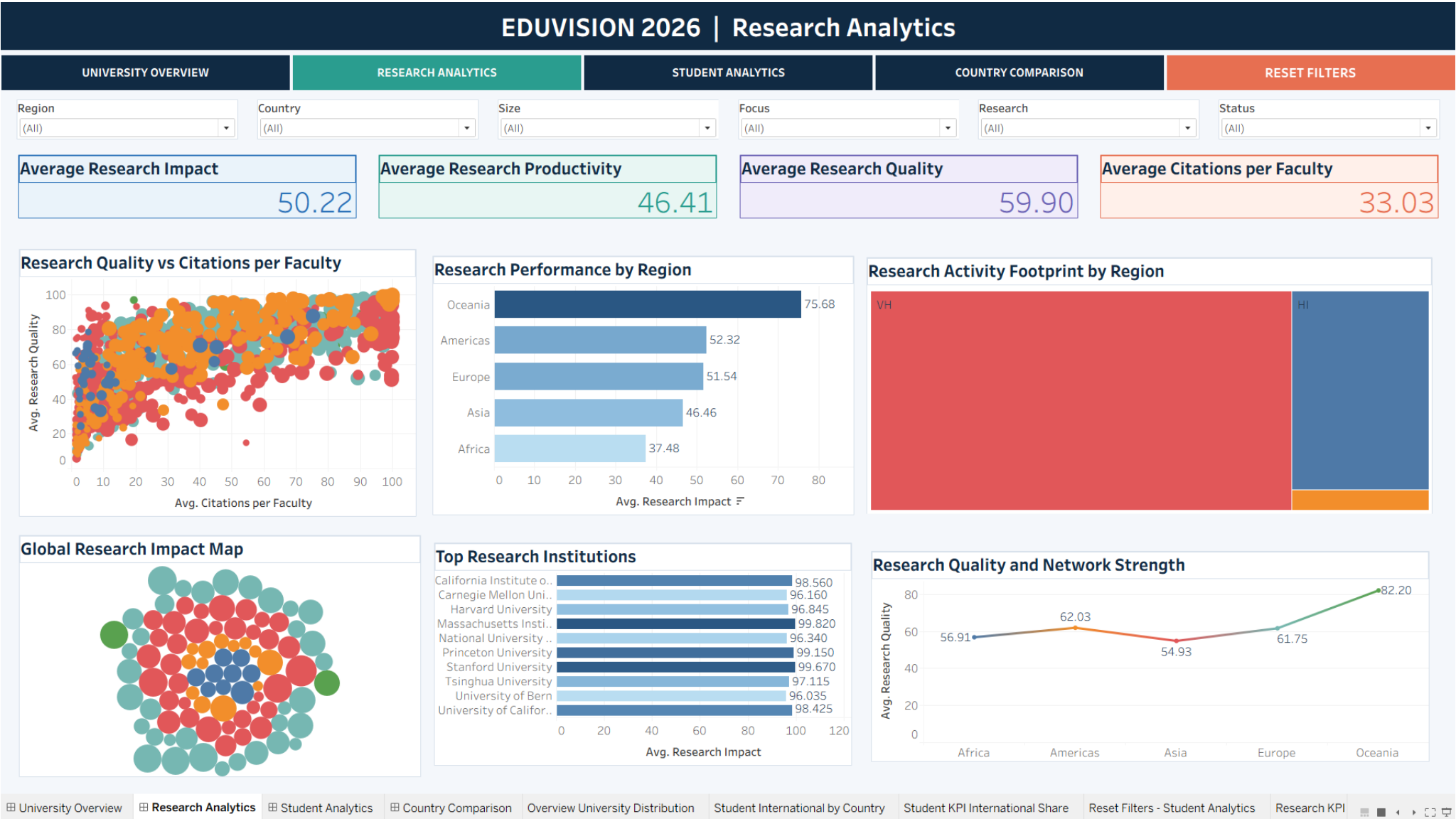


Figure: Completed EduVision_DV Research Analytics dashboard (supplied screenshot).

9. Dashboard 9: Student Analytics

The Student Analytics dashboard interprets the learner and internationalization dimension of institutional performance. It connects student population scale, staff relationship, international students and diversity into a single analytical view.

Displayed KPI cards

KPI	Displayed value
Average Student Population	24,555
Average Students to Staff	17.31
International Students %	14.57
Average Student Diversity	36.60

Visual components

- Student Support vs International Mix (scatter plot using Avg. Students to Staff and Avg. International Students %).
- International Students by Country (selected-country horizontal bars).
- Institution Size Mix (pie chart across L, M, S and XL).
- International Student and Diversity Map (packed-bubble style visualization).
- Student Diversity Benchmark by Region (regional horizontal bars).
- Diversity by Institution Size and Region (band/area-style view).

Interpretation from the supplied dashboard view

- The displayed overall International Students % KPI is 14.57 and the overall Student Diversity KPI is 36.60.
- Oceania leads the displayed regional Student Diversity benchmark at 78.09, followed by Europe 51.76, the Americas 31.96, Asia 22.56 and Africa 13.53.
- Northern Cyprus has the highest displayed country-level International Students value at 72.00%; Luxembourg is next at 52.00%, followed by the UAE at 49.29%.
- The supplied chart also shows Australia 31.03%, Austria 34.67%, Netherlands 31.42%, Qatar 38.00%, Singapore 31.00%, Switzerland 35.67% and United Kingdom 37.31%.
- The scatter plot is designed to examine student-support intensity versus international mix; the visible point cloud suggests considerable institutional dispersion rather than one uniform pattern.

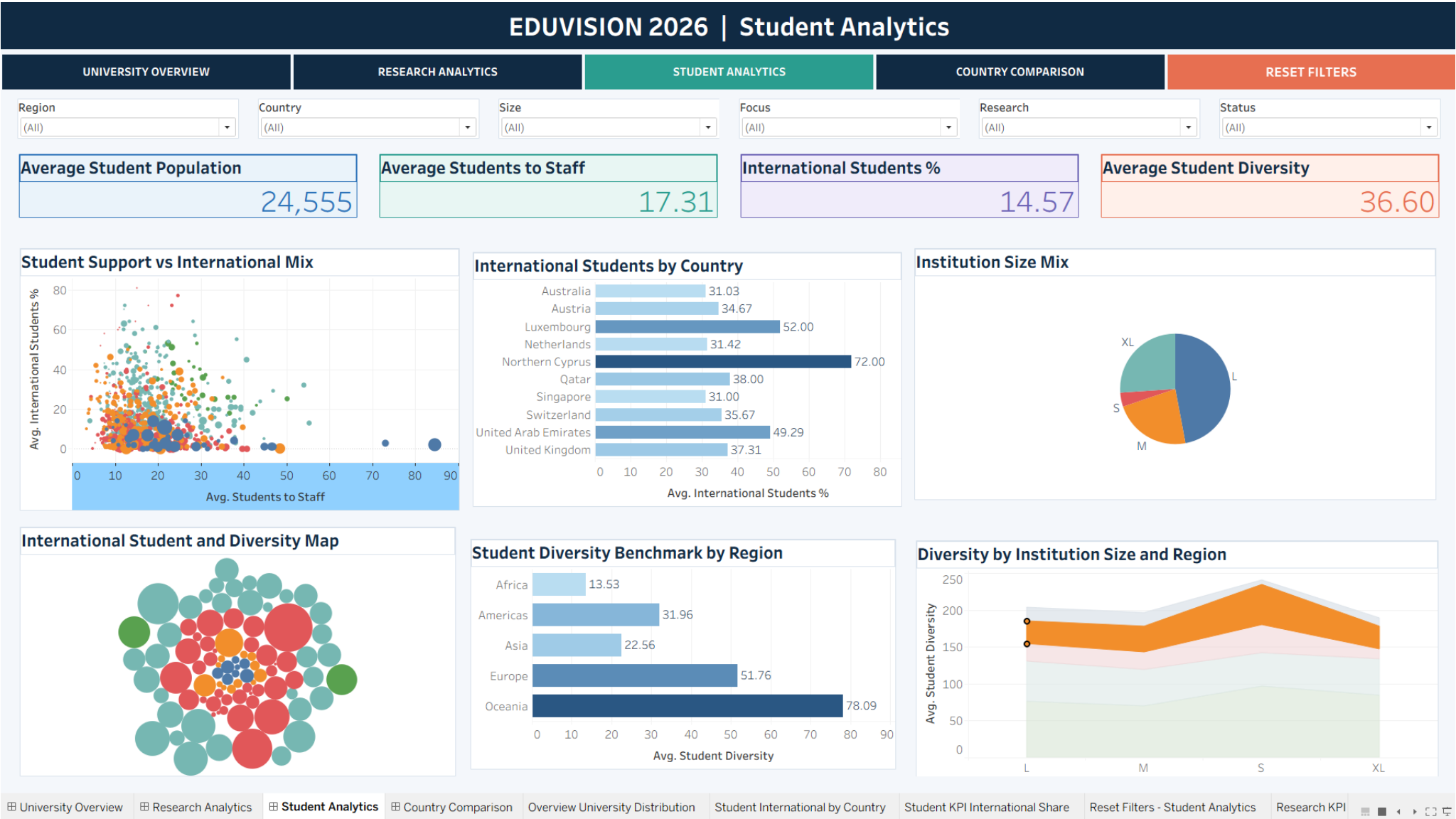


Figure: Completed EduVision_DV Student Analytics dashboard (supplied screenshot).

10. Dashboard 10: Country Comparison

Country Comparison shifts the unit of analysis from individual institutions to national education ecosystems. It is designed for benchmarking selected countries on performance, movement, employer reputation and institutional-status composition.

Displayed KPI cards

KPI	Displayed value
Country Count	93
University Count	1,120
Improved Universities	389
Declined Universities	428

Visual components

- Selected Countries: Core Performance (Avg. Global Score).
- University Share by Institution Status (Public, Private for Profit, Private not for Profit).
- Country QS Rank Movement (Positive = Improvement).
- Country Performance Benchmark Map (packed-bubble style view).
- Selected Countries: Employability and Impact (Avg. Employer Reputation).
- Country Contribution Treemap.

Interpretation from the supplied dashboard view

- Singapore has the highest displayed selected-country Core Performance score at 99.15, followed by Qatar 90.97 and Netherlands 90.80.
- Qatar shows the highest positive displayed QS rank movement at +10.00; Netherlands and Denmark follow at +7.92 and +7.00.
- Luxembourg and Switzerland show the largest displayed declines at -26.00 and -25.00 respectively.
- Singapore also leads the displayed Employer Reputation measure at 94.80; Qatar is next at 56.50 and Netherlands at 49.82.
- The country status pie communicates composition across Public, Private for Profit and Private not for Profit institutions; exact percentages are not printed in the supplied screenshot and therefore are not numerically asserted here.

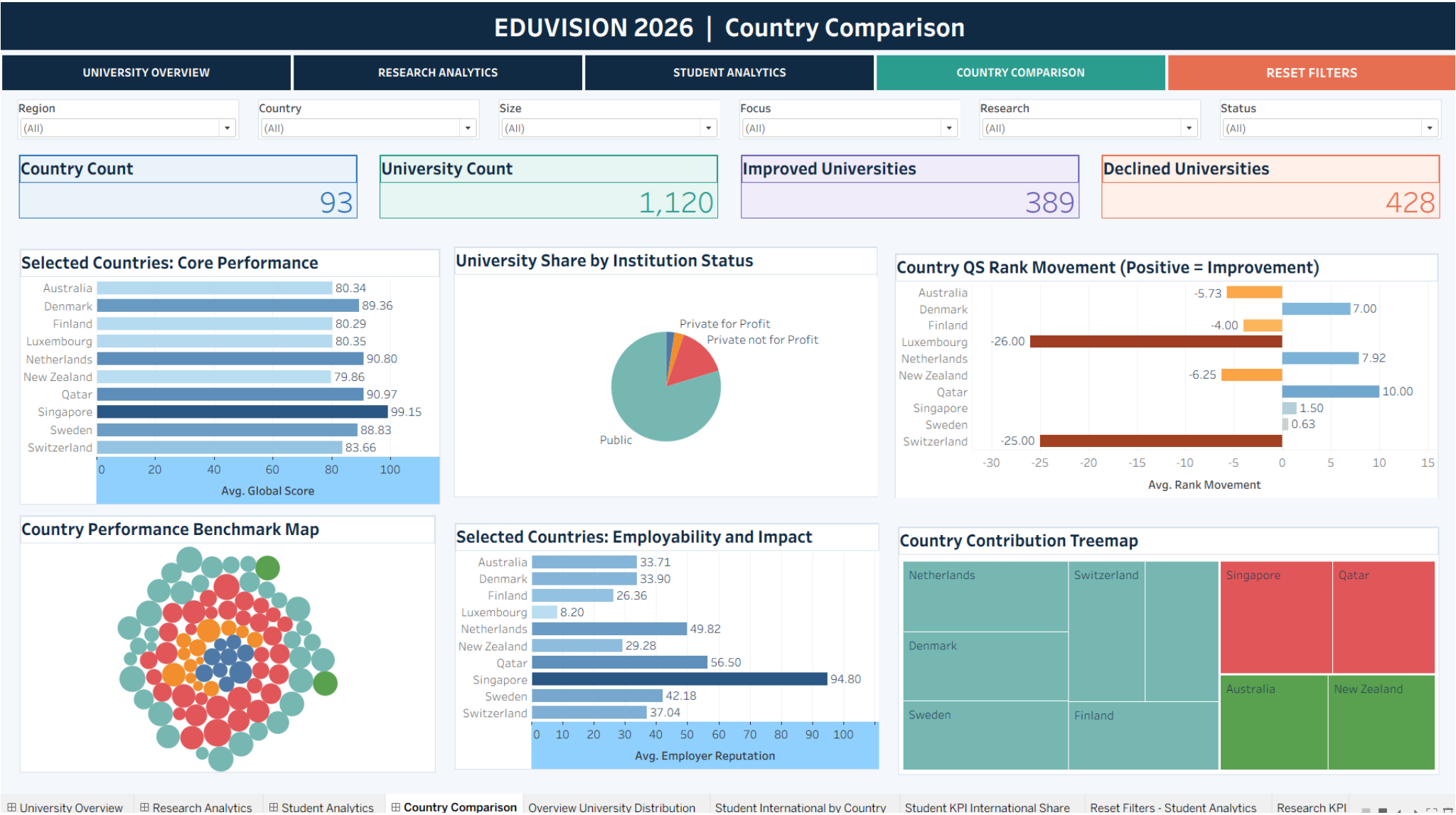


Figure: Completed EduVision_DV Country Comparison dashboard (supplied screenshot).

11. Cross-Dashboard Analysis and Interpretation

Portfolio-Level Signals Visible Across the Dashboards

Signal	Evidence visible in dashboard	Interpretation boundary
Oceania as a displayed benchmark leader	Oceania records the highest shown regional average Global Score (80.25), Research Impact (75.68), Research Quality (82.20) and Student Diversity (78.09).	This is a descriptive cross-dashboard observation from the supplied views, not a causal conclusion.
Africa as the lowest displayed benchmark on several measures	Africa records the lowest shown regional Global Score (42.11), Research Impact (37.48), and Student Diversity (13.53), and the largest negative regional QS rank movement (-29.77).	The pattern is consistent across multiple dashboard views, but the dashboards do not establish why the pattern occurs.
Country momentum is uneven	The selected-country view shows gains for Qatar (+10.00), Netherlands (+7.92) and Denmark (+7.00), while Luxembourg (-26.00) and Switzerland (-25.00) show large declines.	Rank movement should be interpreted separately from absolute performance.
Research quality and citations move together visually	The research scatter plot displays a broadly upward relationship between citations per faculty and research quality.	This is a visual association within the plotted data, not proof of causation.
Top institutions are closely clustered near the score ceiling	The University Overview Top-10 values are all close to 100, while the Research Top-10 values are also tightly grouped in the high 90s.	This means small numerical differences can materially affect ordering among elite institutions.
Internationalization is not uniform	The Student Analytics country chart ranges from around 31% to 72% across the displayed selected countries, while regional diversity varies from 13.53 in Africa to 78.09 in Oceania.	The dashboard reveals concentration and dispersion that merit segmented interpretation.

Decision-Support Perspective

- Students can compare global score, reputation, employer outcomes, student environment and internationalization.
- Researchers can examine research impact, productivity, quality, citations and regional differences.
- Administrators can benchmark their institutions against regions, countries and peer groups.
- Policymakers can examine country-level performance and movement patterns.
- Education consultants can use the filterable views to create segmented comparisons rather than one-size-fits-all recommendations.

12. Testing, Validation and Delivery

Testing Activities Defined in the Project

- Validate KPI calculations.
- Verify ranking calculations.
- Test dashboard interactions.
- Validate educational metrics.

Project-Specification Evaluation Criteria

- Dataset completeness target: above 95%.
- Missing-data target in the cleaning specification: less than 2%.
- KPI milestone: six or more KPIs generated and correctness verified as part of the development plan.
- Dashboard milestone: four dashboards integrated.
- Delivery milestone: fully documented, portfolio-ready dashboard suite.

Delivery Artifact

The supplied final Tableau packaged workbook is the primary delivery artifact. The workbook is accompanied here by a presentation view and this detailed report for explanation, defense and documentation.

13. Project Outcomes and Business Value

- Integrated a large multi-source higher-education dataset into one analytical structure.
- Converted raw ranking indicators into six defined KPIs.
- Built a four-dashboard decision-support interface rather than a single static ranking table.
- Added cross-filtering, parameters, dashboard actions and reset controls to improve exploration.
- Created distinct views for institutional overview, research, students and country benchmarking.
- Enabled descriptive comparisons that support different stakeholder questions from the same integrated dataset.

Business Value by Perspective

Stakeholder	Question supported	Relevant dashboard capabilities
Student	Which institutions/countries combine strong overall performance with attractive student and international profiles?	University Overview + Student Analytics + Country Comparison
Researcher	Which institutions/regions show strong research impact, quality and citation performance?	Research Analytics + University Overview
Administrator	How does an institution/country compare with peers?	Filters + Top-10 views + Country Comparison
Policymaker	How do regions/countries differ on ranking, research and diversity measures?	Country Comparison + regional charts
Consultant	How can comparisons be tailored to specific client criteria?	Common filters + Metric Selector parameter + multiple dashboards

14. Limitations and Data Interpretation Boundaries

- The final dataset is a merged analytical dataset; source definitions and ranking methodologies remain those of the respective source datasets.
- The project summary does not specify the exact mathematical normalization transform used for QS and THE ranks, so this report does not reconstruct one.
- Some labels in the supplied screenshots are visually truncated, especially long university names in Top-10 tables; those labels are preserved rather than expanded into unverified names.
- Dashboard associations should not automatically be interpreted as causal relationships. For example, the citation-quality scatter suggests association, not causation.
- The country and regional dashboards summarize aggregates. Aggregation can hide institution-level heterogeneity.
- Packed-bubble/treemap views communicate composition and relative size effectively, but precise numeric values may not be printed for every category in the screenshot.
- Rank movement is directionally interpreted by the dashboard convention “Positive = Improvement”; movement and absolute score are complementary, not interchangeable.

15. Proposed Improvements (Future Work)

Status of this section: The items below are proposed improvements only. They are not claimed to be part of the completed EduVision_DV implementation unless already described elsewhere as completed.

Proposed improvement	Rationale
Normalization transparency	Document and expose the exact QS/THE normalization equations and any treatment of missing ranks.
Data lineage panel	Add a dedicated workbook page showing source, refresh date, transformation stages and KPI lineage.
Peer-group benchmarking	Allow users to define peer groups explicitly by region, size, focus or research profile.
Trend analysis	Where consistent multi-year institution-level data are available, add longitudinal trend views for KPI movement.
Statistical diagnostics	Provide correlation coefficients or regression diagnostics alongside

Proposed improvement	Rationale
	visual associations where analytically appropriate.
Usability enhancements	Add tooltip explanations for each KPI, formula help text and clearer definitions for rank movement.
Accessibility	Review color contrast, keyboard navigation and label readability for all dashboard users.
Governance	Add a documented data-quality checklist and version history for future data refreshes.

16. Technical Viva / Defense Notes

How to explain the project in 60 seconds

“EduVision_DV is a completed higher-education analytics project that integrates QS World University Rankings 2026 with Times Higher Education data into a 2026 analytical dataset of 1,120 universities across 93 countries and 50 fields. I cleaned and integrated the sources, normalized ranking measures, engineered six KPIs and built four interconnected Tableau dashboards: University Overview, Research Analytics, Student Analytics and Country Comparison. The workbook uses filters, parameters, calculated fields, dashboard actions, Top-10 logic, ranking movement and reset controls. The goal was to move from a single ranking number to multidimensional decision support for students, researchers, administrators, policymakers and education consultants.”

Likely Defense Questions and Strong Answers

Question	Answer framework
Why integrate QS and THE?	Because each ranking source captures a different institutional-performance ecosystem. EduVision_DV combines normalized rank information with research, student and institutional indicators so the analysis is not dependent on one ranking source alone.
Why create KPIs instead of only showing raw fields?	KPIs convert multiple raw measures into interpretable analytical concepts, enabling consistent comparison across dashboards and stakeholder questions.
Why normalize ranking values?	QS and THE ranks use different source systems and scales. Normalization places them onto a common analytical basis before combining them in the Global Ranking Score.
What is the most important dashboard?	There is no single “most important” dashboard. University Overview is the portfolio entry point, Research Analytics is research-focused, Student Analytics is learner/internationalisation-focused, and Country Comparison supports national benchmarking.
What does positive rank movement mean?	The dashboard explicitly labels positive movement as improvement. It is a directional change metric and should be read alongside absolute performance scores.
What was technically challenging?	A central challenge is creating a consistent analytical layer across multiple ranking datasets and then building a coherent Tableau interaction model on top of the cleaned data.
Can the dashboard prove why a university improved?	No. The dashboard is descriptive and comparative. It can reveal patterns and relationships, but causal explanations require additional analytical designs and evidence.
What would you improve next?	The strongest next step would be greater methodological transparency: publish normalization logic, KPI lineage, data-refresh metadata and richer statistical diagnostics.

Appendix A. KPI Formula Reference

Global Ranking Score: Average of normalized QS and THE rank scores

Research Impact Score: 35% Citations per Faculty + 45% THE Research Quality + 20% Industry Impact

Faculty-to-Student Ratio: 1 / Students-to-Staff Ratio

International Student Percentage: International Students × 100

Academic Reputation Score: QS Academic Reputation Score

Research Productivity Index: 30% Research Quality + 25% Research Environment + 20% Citations per Faculty + 15% International Research Network + 10% Industry Impact

Appendix B. Visible Dashboard Data Tables

B1. University Overview - Regional Global Score

Region	Displayed Avg. Global Score
Africa	42.11
Americas	59.73
Asia	50.73
Europe	61.27
Oceania	80.25

B2. University Overview - Top University Global Scores

Displayed institution label	Displayed Avg. Global Score
California Institute o...	99.5400
ETH Zurich (Swiss Fe...	99.5550
Harvard University	99.7400
Imperial College Lon...	99.8050
Massachusetts Insti...	99.9750
National University ...	99.3800
Princeton University	99.2600
Stanford University	99.8400
University of Cambri...	99.7500
University of Oxford	99.8950

B3. University Overview - QS Rank Movement by Region

Region	Displayed Avg. Rank Movement
Africa	-29.77
Americas	-12.63
Asia	+4.47
Europe	-17.04
Oceania	-5.83

B4. Research Analytics - Regional Research Impact

Region	Displayed Avg. Research Impact
Oceania	75.68
Americas	52.32
Europe	51.54
Asia	46.46
Africa	37.48

B5. Research Analytics - Research Quality by Region

Region	Displayed Avg. Research Quality
Africa	56.91
Americas	62.03
Asia	54.93
Europe	61.75
Oceania	82.20

B6. Research Analytics - Top Research Institutions

Displayed institution label	Displayed Avg. Research Impact
California Institute o...	98.560
Carnegie Mellon Uni...	96.160
Harvard University	96.845
Massachusetts Insti...	99.820
National University ...	96.340
Princeton University	99.150
Stanford University	99.670
Tsinghua University	97.115
University of Bern	96.035
University of Califor...	98.425

B7. Student Analytics - International Students by Country

Country	Displayed Avg. International Students %
Australia	31.03
Austria	34.67
Luxembourg	52.00
Netherlands	31.42
Northern Cyprus	72.00
Qatar	38.00
Singapore	31.00
Switzerland	35.67
United Arab Emirates	49.29
United Kingdom	37.31

B8. Student Analytics - Student Diversity by Region

Region	Displayed Avg. Student Diversity
Africa	13.53
Americas	31.96
Asia	22.56
Europe	51.76
Oceania	78.09

B9. Country Comparison - Core Performance, Rank Movement and Employer Reputation

Country	Avg. Global Score	Avg. Rank Movement	Avg. Employer Reputation
Australia	80.34	-5.73	33.71
Denmark	89.36	+7.00	33.90
Finland	80.29	-4.00	26.36
Luxembourg	80.35	-26.00	8.20
Netherlands	90.80	+7.92	49.82
New Zealand	79.86	-6.25	29.28
Qatar	90.97	+10.00	56.50
Singapore	99.15	+1.50	94.80
Sweden	88.83	+0.63	42.18
Switzerland	83.66	-25.00	37.04

Visual-only elements: The supplied screenshots also contain pie/treemap/packed-bubble components for which exact numeric values are not printed. Those visual elements are included in the dashboard screenshots in Appendix D and discussed qualitatively in the dashboard sections.

Appendix C. Field Inventory and Technology Stack

#	Field	Category
1	Rank QS	Identity / segmentation
2	Previous Rank	Identity / segmentation
3	Name	Identity / segmentation
4	Region	Identity / segmentation
5	Size	Identity / segmentation
6	Focus	Identity / segmentation
7	Research	Identity / segmentation
8	Status	Identity / segmentation
9	Academic Reputation Score	QS / institutional
10	Academic Reputation Rank	QS / institutional
11	Employer Reputation Score	QS / institutional
12	Employer Reputation Rank	QS / institutional
13	Faculty Student Ratio Score	QS / institutional
14	Faculty Student Ratio Rank	QS / institutional
15	Citations per Faculty Score	QS / institutional
16	Citations per Faculty Rank	QS / institutional
17	International Faculty Score	QS / institutional
18	International Faculty Rank	QS / institutional
19	International Student Score	QS / institutional
20	International Student Rank	QS / institutional
21	International Students Diversity Score	QS / institutional
22	International Students Diversity Rank	QS / institutional
23	International Research Network Score	QS / institutional
24	International Research Network Rank	QS / institutional

#	Field	Category
25	Employment_Outcomes_Score	QS / institutional
26	Employment_Outcomes_Rank	QS / institutional
27	Sustainability_Score	QS / institutional
28	Sustainability_Rank	QS / institutional
29	QS_Overall_Score	QS / institutional
30	Rank_THE	THE / institutional
31	Student_Population	THE / institutional
32	Students_to_Staff_Ratio	THE / institutional
33	International_Students	THE / institutional
34	Female_to_Male_Ratio	THE / institutional
35	THE_Overall_Score	THE / institutional
36	Teaching	THE / institutional
37	Research_Environment	THE / institutional
38	Research_Quality	THE / institutional
39	Industry_Impact	THE / institutional
40	International_Outlook	THE / institutional
41	Year	Identity / segmentation
42	Country	Identity / segmentation
43	QS_Rank_Normalized	Derived KPI / normalized
44	THE_Rank_Normalized	Derived KPI / normalized
45	KPI_Global_Ranking_Score	Derived KPI / normalized
46	KPI_Research_Impact_Score	Derived KPI / normalized
47	KPI_Faculty_to_Student_Ratio	Derived KPI / normalized
48	KPI_International_Student_Percentage	Derived KPI / normalized
49	KPI_Academic_Reputation_Score	Derived KPI / normalized
50	KPI_Research_Productivity_Index	Derived KPI / normalized

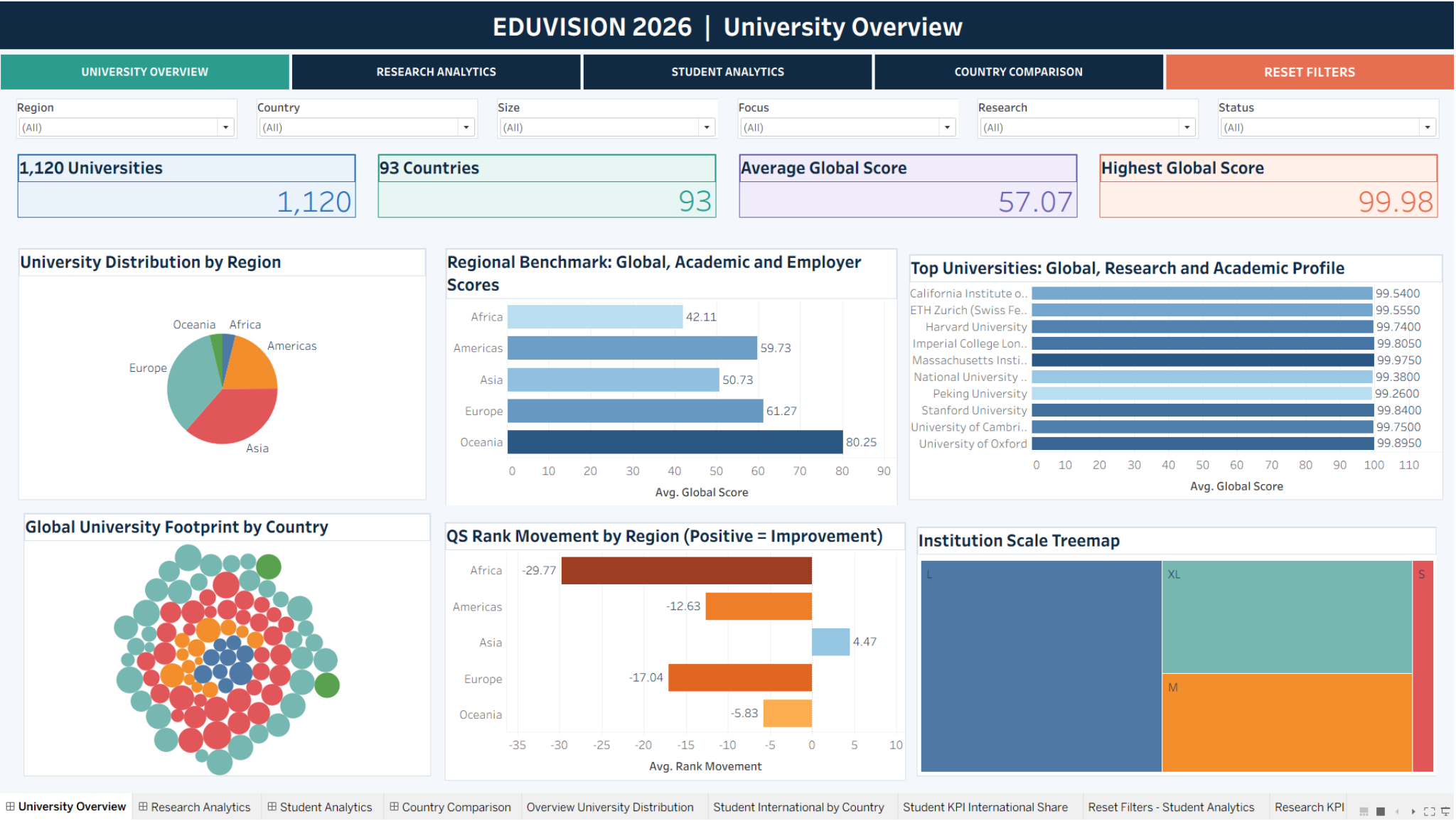
Technology Stack from the Project Specification

Area	Tools / Libraries
Data collection	Python, QS Rankings, World University Rankings
Data processing	Pandas, NumPy
Data cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub

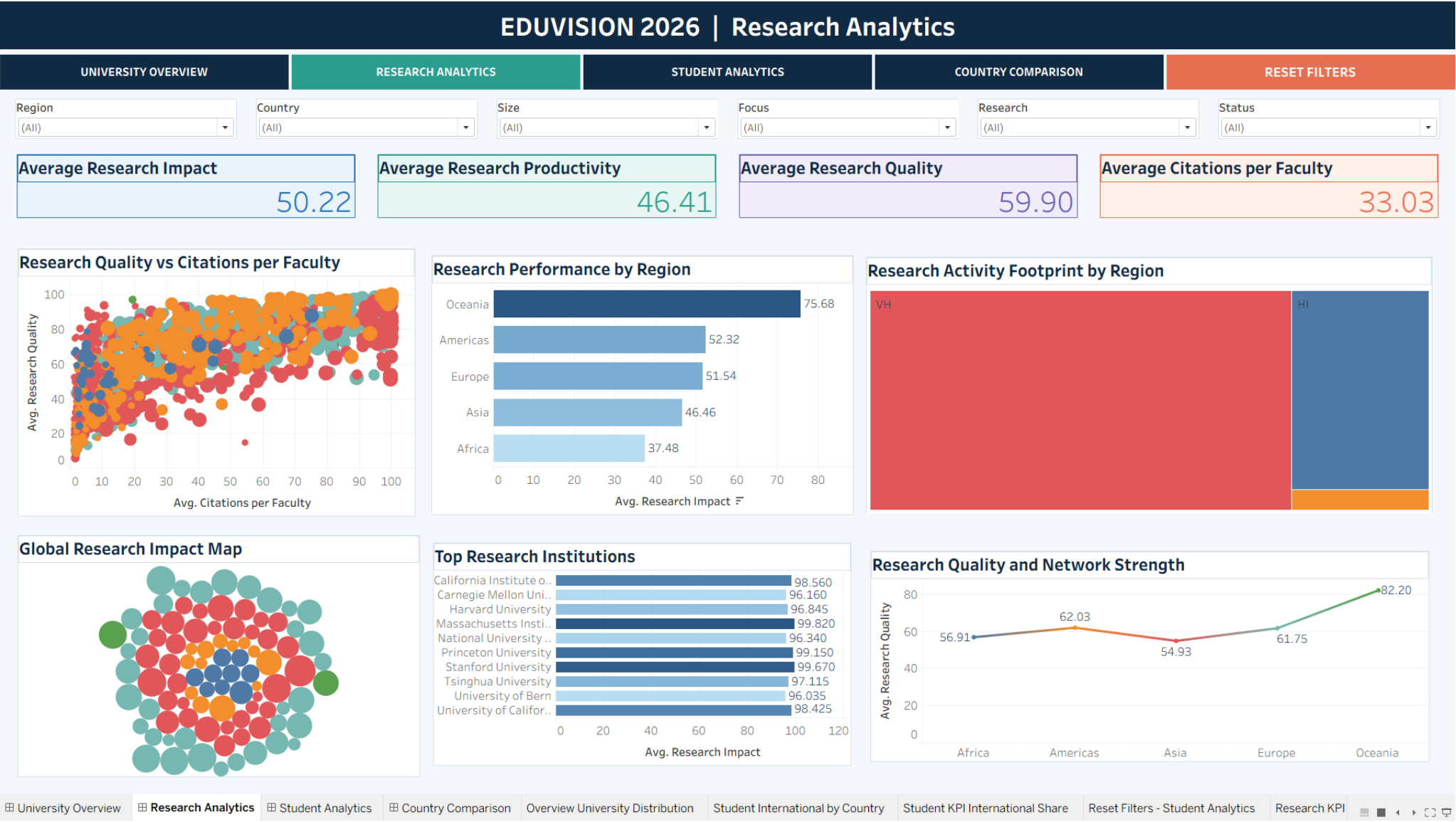
Appendix D. Dashboard Screenshots

The following pages reproduce the supplied completed dashboard screenshots at readable landscape scale. They are included as evidence of the final visualization layer.

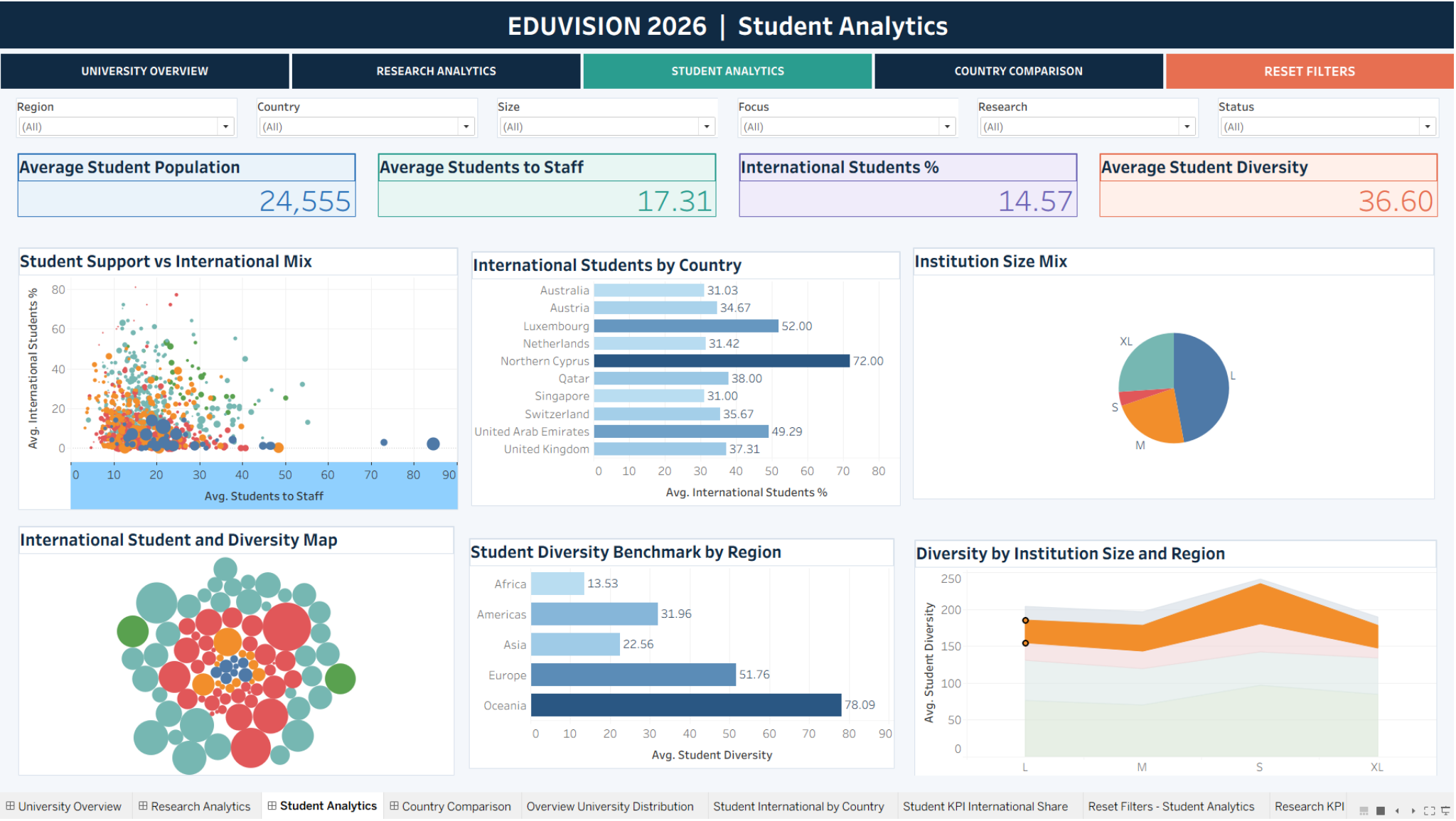
D1. University Overview



D2. Research Analytics



D3. Student Analytics



D4. Country Comparison

